

A Framework of Intrusion Detection System based on Bayesian Network in IoT

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Abstract—The increasing popularity of Internet of Things (IoT) technology has created great influence on human's production mode and life quality. At the same time, the issue of Internet of Things security problem has also become the focus of attention. There are many aspects of IoT security issues. In this paper, we do some research on the problem of network intrusion detection. First, an intrusion detection dataset named UNSW-NB15 is selected as the research object. Then we preprocess the dataset and accomplish the feature selection job to obtain a suitable subset. After the above steps finished, a Bayesian model is built according to the K2 structure learning algorithm. And the parameters are obtained through the Maximum Likelihood Estimation algorithm. Finally, input the testing dataset to classify. The simulation result shows that the system can detect the anomaly intrusion effectively.

Keywords—Internet of Things, Intrusion Detection System, Bayesian Network, UNSW-NB15

I. INTRODUCTION

Modern technologies, especially Internet of Things, have revolutionized human society and they are now coming to a white-hot development stage. With the rapid development of the Internet of Things technology, more and more security problems are emerging. The Internet of Things is vulnerable to a variety of attacks, such as application attacks, network attacks, and physical attacks. There is no doubt that the variety of attack is getting richer and quantity of attack is becoming larger even out of what we can control [1]. Moreover, in the era of net interconnection and even the interconnection of things, the security problem is magnified. Usually, once a device or a net is polluted, the others are often spared which will cause significant losses [2]. So in this case, how to establish an effective security system, which aims to detect various intrusions and malicious attacks and ensure the reliability of the Internet of Things, is especially important.

Generally speaking, the steps to build an intrusion detection system (IDS) include the collection of dataset, data cleaning, feature selection of original data, model establishment and classification [3]. The collection and cleaning job can be regarded as the preprocessing task, which is crucial to the latter steps. And feature selection is often a way that improves real-time performance while meeting accuracy requirement. Modeling is a process of data fitting that describes the system. All of the above work is to pave the way for the final decision.

From another perspective, intrusion detection can be seen as a problem of uncertainty event reasoning under the

condition of known information. As we all know, Bayesian Network (BN), first proposed by Judea Pearl in 1985 [4], is a model based on probability theory and graph theory for uncertainty expression and probabilistic reasoning. It is widely used in various fields such as fault detection, industrial control, data mining, medical diagnosis, and military.

In this paper, a framework of IDS using the Bayesian Network and UNSW-NB15 dataset is proposed. The rest of this paper is organized as follows. Section II provides a summary of related work in IDS. Section III describes some basic theory of the intrusion detection under IoT and some knowledge about Bayesian network. In Section IV, the details of the whole IDS will be explained in the order of data preprocessing, feature selection, Bayesian structure & parameter learning and classification decision. Section V gives a simulation and we conclude in Section VI.

II. RELATED WORK

A lot of research has been done on the IDS. Since the new data set UNSW-NB15 is chosen in this paper, the following work is related to it.

In [5], the value of the center point for each feature is first calculated, and Association Rule Mining (ARM) is used to select the best subset. The Expectation Maximization (EM) clustering, Logistic Regression and Naïve Bayes are used as the decision engine for intrusion detection with the accuracy 77.2%, 83.0% and 79.5% respectively. In [6], Based on the idea of binary tree, a hybrid detection framework is constructed. For the known attack, use the best model for each type, while for the unknown, classify it into normal or anomaly. However, the complexity of this method is too high and the practicality is not high. In [7], the author analyzed the KDD'99 and UNSW-NB15 from the distribution of data and the similarity of training and testing data. Five different classifiers are used to prove that the UNSW-NB15 is more complex and has more modern network features. In [8], Artificial Neural Networks is used for detection of Dos. In order to improve the detection accuracy, the multi-layer perceptron is used and Dos detection accuracy can reach 97%. In [9], data is first divided into three categories according to different network protocols (TCP, UDP and others), and based on the Information Gain, at least 12 features are selected and Decision Tree is to make decision. In [10], a Random Forest with 800 trees is proposed for IDS, where the parameters are obtained by the grid search method. In [11], a method using Genetic algorithm and Support Vector Machine is used for IDS. The true positive rate (TPR),

false positive rate (FPR) and the number of selected features are used as the parameter construction fitness function in the genetic algorithm. In [12], an three-layer IDS framework is proposed where ARM is used to generate the strongest feature subset. Naive Bayes and EM clustering are used as the decision engine to classify. In [13], A variety of machine learning and data mining algorithms are used to for feature selection on the UNSW-NB15 dataset. A Random Forest classifier is used with 5 attributes and the result is better than that in [12]. In [14], five different strategies to select the best subset are used and J48 classifier with GR filter can achieve the highest 88% accuracy with 18 features.

III. BASIC THEORY

A. Network Intrusion Detection System of IoT

A common IoT security framework is shown in Fig. 1 [15]. This framework contains 3 parts, perception layer, transportation layer as well as application layer. A large amount of data pass through the transport layer, which makes it extremely easy to be attacked. Therefore, an effective network intrusion detection system is critical to the entire IoT security system.

In fact, the IDS system can be partitioned from a number of different perspectives. The first one is network-based IDS or host-based IDS [16]. The network-based IDS refers to detect an activity through analysis of the network packets. The host-based IDS means deploying the intrusion detection software or program on the corresponding host, which aims to analyze the system log. From another view, intrusion detection systems can be divided into misuse detection and anomaly detection [17]. The former one can tell you the exact type of intrusion attack. But it fails in detecting unknown attacks. The result of the other detection system is relatively simple, that is, normal or abnormal. It is constantly receiving attention due to the ability to identify some new attacks. In this paper, a network-based anomalies system will be introduced to solve the intrusion detection problem.

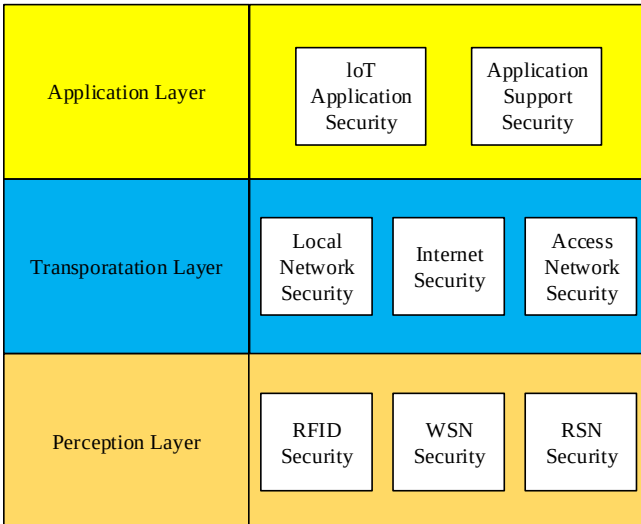


Fig. 1. A common IoT security framework

B. Bayesian Network

In the process of simulating the uncertainty representation, reasoning and learning, the Bayesian network provides a method to visualize the knowledge diagram. BN is a directed acyclic graph consisting of two parts. Part of the

qualitative information, that is, the network structure, uses nodes and edges to represent the reality of things and their associations; The other part, that is, network parameters, represents the quantity of correlation degree between variables through the conditional probability table [18].

In the Bayesian network, nodes without direct connection are conditionally independent. In this case, the joint probability can be decomposed into the following formula:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | Pa(X_i)) \quad (1)$$

Conditional independence greatly reduces the complexity of the probabilistic model in the computation process. Assuming all the variables in the model 'Rain' have two different states, then the joint probability will be as follow:

$$P(C, S, R, W) = P(C) * P(S | C) * P(R | C) * P(W | S, R) \quad (2)$$

Joint probability can be obtained by only 9 independent probabilities. Compared with the original, conditional independence can greatly reduce the computational complexity of Bayesian networks.

Bayesian network inference refers to the calculation of posterior probabilities of interested variables after giving evidence variables according to Bayes formula:

$$P(Y | X) = \frac{P(X, Y)}{P(X)} \quad (3)$$

IV. PROPOSED INTRUSION DETECTION SYSTEM

In this section, a framework for intrusion detection using Bayesian network is proposed as shown in Fig. 2.

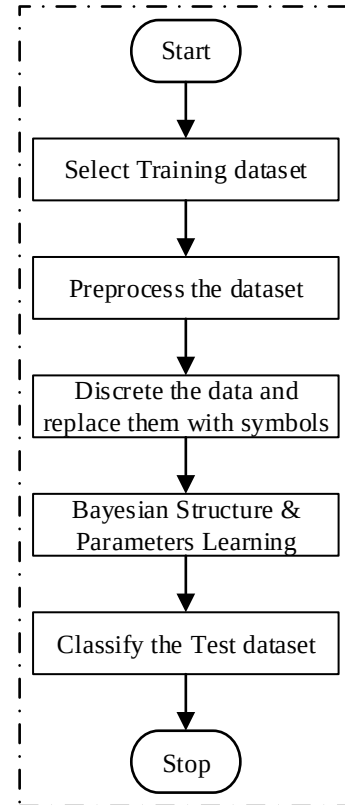


Fig. 2. Framework for IDS using Bayesian Network

A. Dataset

When evaluating the performance of the NIDS system, it is important to have a complex data set that can reflect the normal or attack situations of the real network [19]. There are many scholars and research institutions that have done a lot of work on the dataset of network intrusion detection. Among them, KDD'99 and its updated version NSLKDD are classical [20]. However, redundancy and data imbalance exist in them. Besides, with the development of network technology, many new types of attacks have emerged.

TABLE I. UNSW-NB15 DATASET

Number	Description	Number	Description
1	dur	23	dwin
2	proto	24	tcprtt
3	service	25	synack
4	state	26	ackdat
5	spkts	27	smean
6	dpkts	28	dmean
7	sbytes	29	trans_depth
8	dbytes	30	response_body_len
9	rate	31	ct_srv_src
10	sttl	32	ct_state_ttl
11	dttl	33	ct_dst_ltm
12	sload	34	ct_src_dport_ltm
13	dload	35	ct_dst_sport_ltm
14	sloss	36	ct_dst_src_ltm
15	dloss	37	is_ftp_login
16	sinpkt	38	ct_ftp_cmd
17	dinpkt	39	ct_flw_http_mthd
18	sjit	40	ct_src_ltm
19	djit	41	ct_srv_dst
20	swin	42	is_sm_ips_ports
21	stcpb	43	attack_cat
22	dtcpb	44	label

A complex and balanced data set is especially important for intrusion detection. The UNSW-NB15, created by Moustafa et al [21], is a comprehensive network-based dataset which can reflect the modern network traffic scenarios. It contains 49 different features, which can be divided into 5 parts, that is Flow features, Basic features, Content features, Time features and Additional generated features. The official website provides a pair of training and testing dataset. The number of records in the training set is 82,332 records, where the normal percentage is 45% and anomaly is 55%. The testing set is 175,341 records, where the normal percentage is 32% and abnormal is 68%.

Compared to other data sets, the UNSW-NB15 data set has the following advantages:

- (1) Rich data types, more in line with the characteristics of modern network traffic data;
- (2) The distribution in training and testing dataset is similar.

So, in this paper, we adopt the UNSW-NB15 dataset to do the research on the intrusion detection.

B. Feature selection

In general, dataset contains too many features that are not only unimportant and redundant, but also harmful to the accuracy of the results [22]. The more features in dataset, the more difficult the problem is to solve in modeling and testing. There are 44 features in the UNSW-NB15 data set. Obviously, it is necessary to streamline it to simplify the construction of the model and increase the speed of the

operation. However, the feature selection step not only relies on technology but also requires much experience.

In this paper, the feature selection method is based on the work of [13], where a few different kinds of machine learning techniques are adopted to select the subset. In [13], The selected subset can improve Kappa statistic which means better intrusion detection rates than the one in [12]. So we adopt these features as the research object, *service, sbytes, sttl, smean and ct_dst_sport_ltm*.

C. Bayesian Network Structure Learning

Bayesian network structure learning is based on the premise of a given set of data samples, to analyze a network structure that matches the training sample set and preferably reflects the dependencies between nodes. However, the process to find an optimal structure is a NP- hard problem [23]. Therefore, learning Bayesian network structure from data has always been a research hotspot. Generally speaking, structure learning methods are mainly divided into two categories, one based on the scoring function and the other based on the constraint learning method [24].

In this paper, K2 algorithm presented by Cooper [25] is used to learn the BN. The specific process is as follows:

TABLE II. K2 STRUCTURE ALGORITHM

K2 structure algorithm
<i>Input</i> : $X = \{x_i \mid i = 1, \dots, N\}$,a set of variables, an ordering on the nodes, an upper bound u on the number of parents for each node, and a database D containing m cases. <i>Output</i> : A dag with oriented arcs. <i>Process</i> : for $i = 1$ to N do $\pi_i = \emptyset$ $P_{old} = g(i, \pi_i)$ OKToProceed = ture while OKToProceed and $\pi_i < u$ do let z be the node in $Pred(x_i) - \pi_i$ that maximizes $g(i, \pi_i \cup \{z\})$ $P_{new} = g(i, \pi_i \cup \{z\})$ if $P_{new} > P_{old}$ then $P_{new} = P_{old}$ $\pi_i = (\pi_i \cup \{z\})$ else OKToProceed=false end if end while write('Node', x_i, 'Parents of thie node', π_i) end for end K2

D. Bayesian Network Parameters Learning

Parameter learning refers to learning the quantitative relationship of Bayesian networks based on data, that is, conditional probability.

In the case of relatively complete data, Maximum Likelihood Estimation (MLE) algorithm can be used for parameter learning [26]. The method judges the degree of fitting of the sample and the model according to the likelihood of the sample and the parameter, so that the

parameter with the largest value of the likelihood function is the parameter learned by the Bayesian network.

$$\theta^* = \arg \max_{\theta} L(\theta | D) \quad (4)$$

Suppose a Bayesian network with n variables, $X = \{x_1, x_2, \dots, x_n\}$, where x_i has r_i values and the amount of parent nodes' combination is q_i . With the definition, θ could be $\theta = \{\theta_{i,j,k} | i=1, \dots, n; j=1, \dots, q_i; k=1, \dots, r_i\}$, where the $\theta_{i,j,k} = P(x_i = k | \text{parent}(x_i) = j)$.

So the required logarithmic parameter can be written as:

$$\begin{aligned} l(\theta | D) &= \log L(\theta | D) = \log \prod_{i=1}^m P(d_i | \theta) \\ &= \sum_{i=1}^n \sum_{j=1}^{q_i} \sum_{k=1}^{r_i} m_{i,j,k} \log \theta_{i,j,k} \end{aligned} \quad (5)$$

E. Bayesian Network Inference

The Variable Elimination (VE) algorithm is an easy-to-understand computational simple reasoning algorithm proposed by Zhang et al [27]. Variable Elimination algorithm adopts a straightforward way to decompose the joint probability according to the chain product rules and conditional independence. It is easier to implement by calculating the posterior probability on the BN, but the reasoning result is disposable for one node.

Assuming there is a BN, where Obs are the observed nodes and X is the target node, the posterior probability can be written as

$$P(X | Obs) = \frac{P(X, Obs)}{P(Obs)} \quad (6)$$

Applying the chain rule with condition independence, equation(5) can be presented below:

$$\begin{aligned} P(X | Obs) &= \frac{\sum_{others} P(X, Obs, others)}{\sum_{others} \sum_X P(X, Obs, others)} \\ &= \frac{\sum_{others} \prod_i^N P(A_i | Pa(A_i))}{\sum_{others} \sum_X \prod_i^N P(A_i | Pa(A_i))} \end{aligned} \quad (7)$$

In equation(6), A_i is the node in BN, while the $Pa(A_i)$ is the parent nodes of A_i . In this way, eliminate irrelevant variables $others$ one by one until $P(X, Obs)$ is known. Finally, compute the posterior probability of X given Obs :

$$P(X | Obs) = \frac{P(X, Obs)}{\sum_X P(X, Obs)} \quad (8)$$

V. SIMULATION

In this section, the UNSW-NB15 dataset is selected to do the experiment. In the feature selection part, the result is based on the work of [13]. And then discrete the original

selected data. Based on the above steps, we work in the MATLAB environment to code which aims to build a Bayesian Model with the Bayes Net Toolbox written by Kevin Murphy [28]. Finally, a Bayesian network classifier is constructed for intrusion detection.

First, based on the former work, the selected features are *attack*, *smean*, *sbytes*, *service*, *ct_dst_sport_ltm*, *sttl*. When it comes to discrete original data, we first analyzed the distribution of the data. Discretization is based on the distribution of data and the association with corresponding categories. Table III is a discretization example of *smean*.

TABLE III. AN EXAMPLE OF DISCRETIZATION FOR FEATURE *smean*

Original	Discrete	Original	Discrete
$smean < 60$	1	$60 < smean < 100$	2
$100 < smean < 200$	3	$smean > 200$	4

Then, we use K2 learning method to obtain a BN structure, as shown in Fig. 3. Here, to make the structure clear and simple as well as to ensure a high performance, an upper bound on the number of parents for each node is set to be 2 after lots of experiments. The prior order as an input is $\{attack, smean, sbytes, service, ct_dst_sport_ltm, sttl\}$.

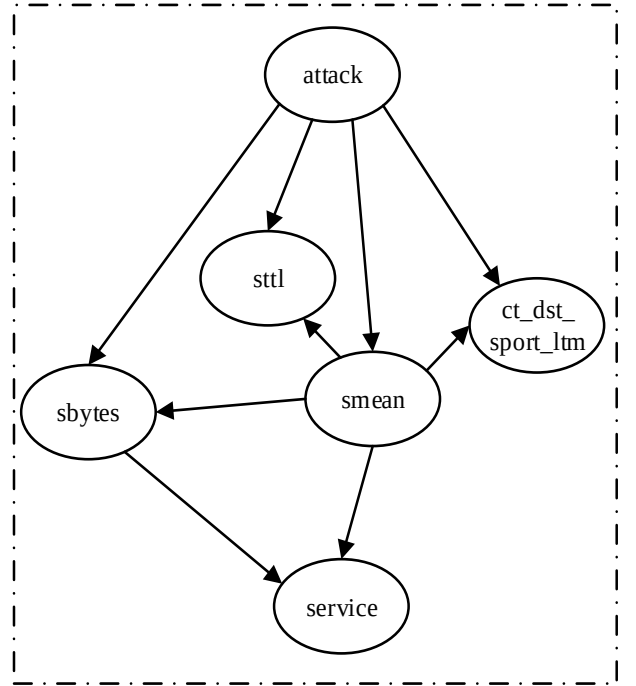


Fig. 3. The learned structure of BN for IDS

TABLE IV. THE CONFUSION MATRIX OF CLASSIFICATION RESULT

	Normal	Attack
Normal	TN = 52106	TP = 95292
Attack	FN = 3894	FP = 24049

For the anomaly detection problem, we first get the confusion matrix of test dataset.

True Negative (TN) means the number of normal examples that are classified correctly.

False Negative (FN) represents the number of normal examples that are regarded as abnormal.

True Positive (TP) stands for the amount of attack ones which are classified as anomaly.

False Positive (FP) is the number of attack records that are wrongly divided into normal part.

With the definition above, to evaluate the performance of a classifier, we adopt the following metrics, the False Alarm Rate (FAR) and the Accuracy.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$FAR = \frac{1}{2} \left(\frac{FP}{FP + TN} + \frac{FN}{FN + TP} \right) \quad (10)$$

The comparison results of classification accuracy and FAR between the proposed framework and those in [13] are separately shown in table V and VI. It can be seen from table V that the FAR both in training dataset and testing dataset are higher than those in [13]. While the accuracy in this paper is a little higher than the one in [13].

TABLE V. COMPARISON OF FAR

FAR	Proposed framework	Framework in [13]
Testing dataset	17.14%	12.45%
Training dataset	17.75%	11.35%

TABLE VI. COMPARISON OF ACCURACY

Accuracy	Proposed framework	Framework in [13]
Training dataset	82.997%	82.992%
Testing dataset	84.046%	81.617%

VI. CONCLUSION

The increasing development of IoT technology creates huge impact on human life. However, many security issues have also emerged. In this paper, the problem of anomaly intrusion detection in the security defense of the Internet of Things transport layer is studied. An interpretable model based on Bayesian network is proposed through structure & parameter learning process, where the UNSW-NB15 data set is used. Simulation results show that the proposed Bayesian network model has a good effect on anomaly detection and has an accuracy of more than 84%. But there are also many shortcomings, and the next work will focus on the selection of a more appropriate subset of features and the promotion of accuracy and reduction of false alarm rates.

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